
Your answers should be i) precise, ii) to-the-point, and iii) with legible handwriting.

1. Answer the following:

- i) In the coordinated MAPF example studied in the class, suppose we modify the problem as follows: i) there is only one drone and multiple trucks in the workspace, ii) the paths of the trucks cannot be modified, iii) the path of the drone can be modified, that is, the drone can move to the trajectory of a truck and ride on it for some distance (for recharging). iv) cost of traversing each edge is 1 unit.

Suppose, given (s^{dr}, g^{dr}) of the drone, it's shortest path is $\pi_{dr} = (e_{dr}^0, e_{dr}^1, \dots, e_{dr}^t)$. It needs to go for recharging at timestep p till q (these are known parameters). For that, it has to find one of the truck trajectories and move there at time p and come back to its initial trajectory π^{dr} at time q . Consider the trajectory of truck i is $\pi_i = (e_i^0, e_i^1, \dots, e_i^{l_i})$.

Write mathematical expression for the assignment of the drone to one of the trucks such that the cost of deviation from π^{dr} is minimized. 4

- ii) Suppose, there are n players. Each player announces a number in the set $\{1 \dots K\}$, where K is a fixed positive integer. A prize of \$1 is split equally between all the people whose number is closest to $\frac{2}{3}$ of the average number. *Formulate* this as a strategic form game. 3

2. Consider the following instance of the Prisoner's Dilemma problem.

		Player-2	
		C	NC
Player-1	C	$(-x, -x)$	$(-x, -3)$
	NC	$(-3, -x)$	$(-6, -6)$

Find the value of x for which

- i) The profile (C, C) is a strongly dominant strategy equilibrium. 2
- ii) The profile (C, C) is a weakly dominant strategy equilibrium, but not a strongly dominant strategy equilibrium 2
- iii) Is (C, C) is a *self-enforcing* strategy profile ? Give reason. 2

3. Answer the following:

- i) Which is/are the *Pareto optimal* outcome(s) in the following game? 3

		Player-2		
		X	Y	Z
Player-1	A	(6, 6)	(8, 20)	(0, 8)
	B	(10, 0)	(5, 5)	(2, 8)
	C	(8, 0)	(20, 0)	(4, 4)

- ii) Mention a pair of outcomes which do not Pareto dominate each other. 2
- iii) Show that the strategy profile (C, Z) is a PSNE of the following game. 2

		Player-2		
		X	Y	Z
Player-1	A	(6, 6)	(8, 20)	(0, 8)
	B	(10, 0)	(5, 5)	(2, 8)
	C	(8, 0)	(20, 0)	(4, 4)